

Inverted Index

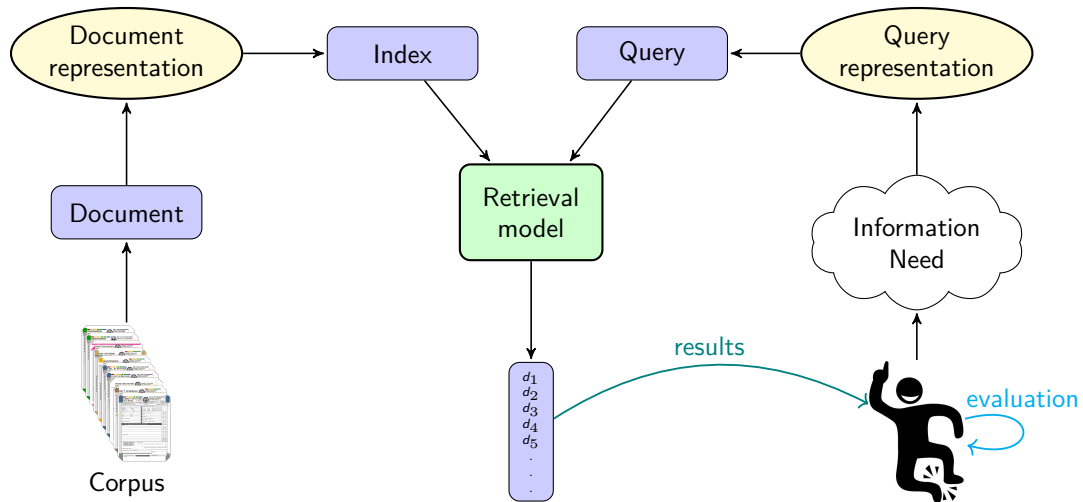
Introduction to Information Retrieval

Dwaipayan Roy

Department of Computational and Data Sciences



Information Retrieval: A Graphical Representation



The basics of **searching** in information retrieval

bag of words approach

- **Information Retrieval**: process of seeking out **textual documents** of *unstructured nature*, that satisfies the *information need* (of an user) from within a *collection of documents*.
 - ▶ **Unstructured nature**: no specific structure (a *collection of words*).
 - ▶ **Information need**: the query of the user (a *set of keywords*).

The basics of **searching** in information retrieval

bag of words approach

- **Information Retrieval**: process of seeking out **textual documents** of *unstructured nature*, that satisfies the *information need* (of an user) from within a *collection of documents*.
 - ▶ **Unstructured nature**: no specific structure (a *collection of words*).
 - ▶ **Information need**: the query of the user (a *set of keywords*).

The task

Given a query retrieve the *relevant* documents.

The basics of **searching** in information retrieval

bag of words approach

- **Information Retrieval**: process of seeking out **textual documents** of *unstructured nature*, that satisfies the *information need* (of an user) from within a *collection of documents*.
 - ▶ **Unstructured nature**: no specific structure (a *collection of words*).
 - ▶ **Information need**: the query of the user (a *set of keywords*).

The task

Given a query retrieve the *relevant* documents.



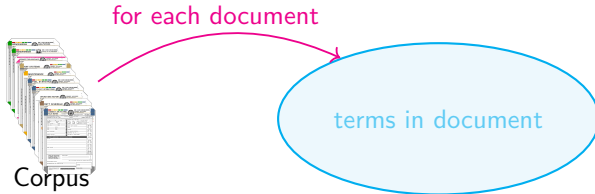
The basics of **searching** in information retrieval

bag of words approach

- **Information Retrieval**: process of seeking out **textual documents** of *unstructured nature*, that satisfies the *information need* (of an user) from within a *collection of documents*.
 - ▶ **Unstructured nature**: no specific structure (a *collection of words*).
 - ▶ **Information need**: the query of the user (a *set of keywords*).

The task

Given a query retrieve the *relevant* documents.



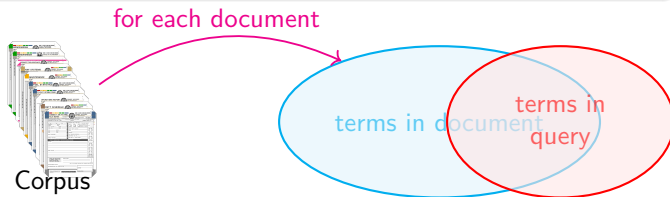
The basics of **searching** in information retrieval

bag of words approach

- **Information Retrieval:** process of seeking out **textual documents** of *unstructured nature*, that satisfies the *information need* (of an user) from within a *collection of documents*.
 - ▶ **Unstructured nature:** no specific structure (a *collection of words*).
 - ▶ **Information need:** the query of the user (a *set of keywords*).

The task

Given a query retrieve the *relevant* documents.



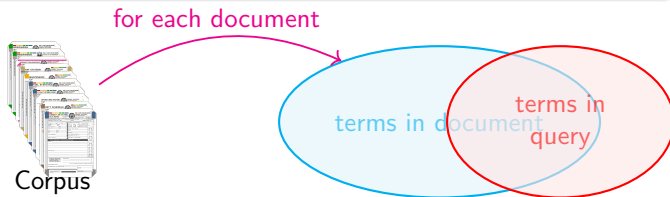
The basics of **searching** in information retrieval

bag of words approach

- **Information Retrieval**: process of seeking out **textual documents** of *unstructured nature*, that satisfies the *information need* (of an user) from within a *collection of documents*.
 - ▶ **Unstructured nature**: no specific structure (a *collection of words*).
 - ▶ **Information need**: the query of the user (a *set of keywords*).

The task

Given a query retrieve the *relevant* documents.



The task

Given a set of terms, retrieve the documents containing those terms.

The basics of **searching** in information retrieval

bag of words approach

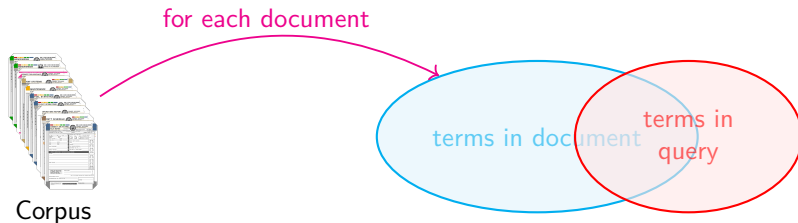


Figure: The bag of words approach to information retrieval.

Retrieval in BOW approach

- Document → list of keywords.
- User's information need → (natural-language) query → list of keywords.

The basics of **searching** in information retrieval

bag of words approach

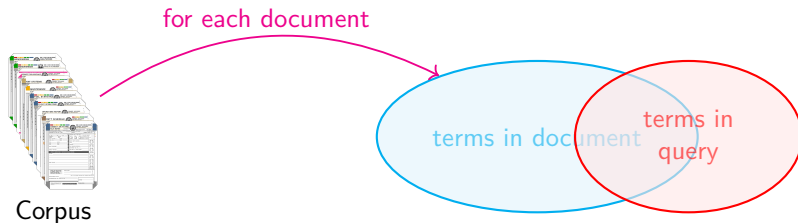
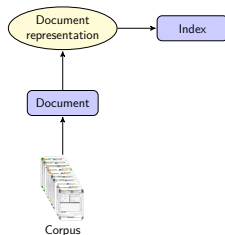


Figure: The bag of words approach to information retrieval.

Retrieval in BOW approach

- Document → list of keywords.
- User's information need → (natural-language) query → list of keywords.
- Measure overlap between query and documents.

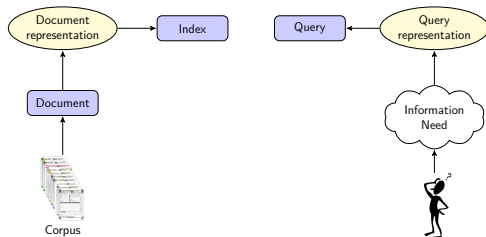
The basics of **searching** in information retrieval



Retrieval in BOW approach

- Document → list of keywords.
- User's information need → (natural-language) query → list of keywords.
- Measure overlap between query and documents.

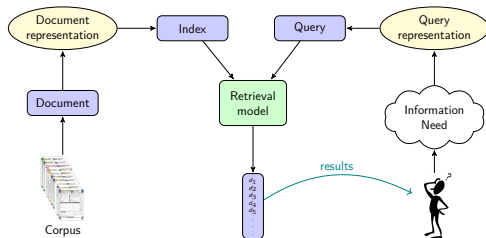
The basics of **searching** in information retrieval



Retrieval in BOW approach

- Document → list of keywords.
- User's information need → (natural-language) query → list of keywords.
- Measure overlap between query and documents.

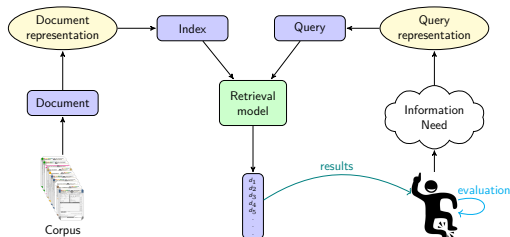
The basics of **searching** in information retrieval



Retrieval in BOW approach

- Document → list of keywords.
- User's information need → (natural-language) query → list of keywords.
- Measure overlap between query and documents.

The basics of **searching** in information retrieval



Retrieval in BOW approach

- Document → list of keywords.
- User's information need → (natural-language) query → list of keywords.
- Measure overlap between query and documents.

The basics of **searching** in information retrieval

bag of words approach

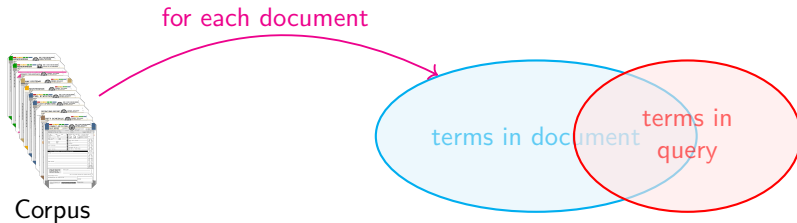


Figure: The bag of words document representation.

The basics of **searching** in information retrieval

bag of words approach

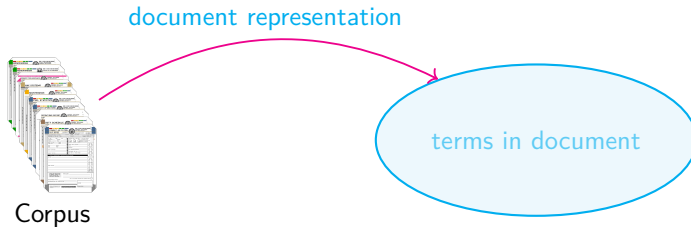


Figure: The bag of words document representation.

Basics of bag-of-words approach

document representation

document representation in bag-of-words approach

- store the documents (**corpus**) such that, given some terms (**query**) we can quickly find only those documents containing those terms.

Basics of bag-of-words approach

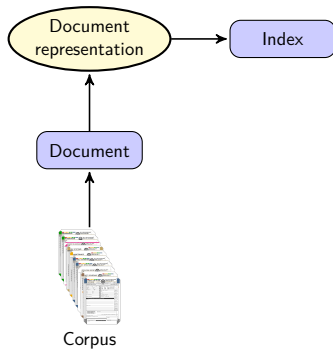
document representation

document representation in bag-of-words approach

- store the documents (**corpus**) such that, given some terms (**query**) we can quickly find only those documents containing those terms.
- Two aspects of document representation:
 - ① selection of terms in the document; (**text processing**)
 - ② storing the terms. (**indexing**)

Document representation

1. text processing

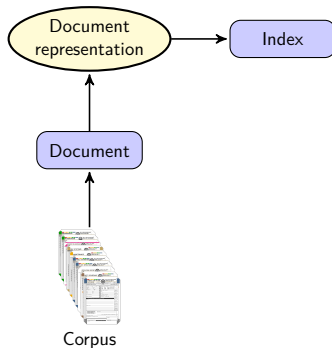


Document representation

- 1 text processing
- 2 indexing

Document representation

1. text processing



Document representation

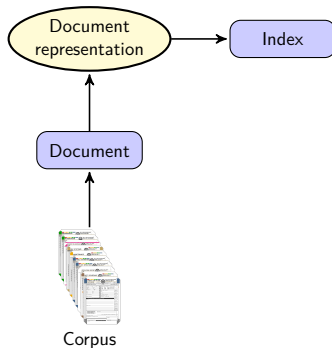
- ① text processing
- ② indexing

Text processing

- a. term selection;
- b. term filtering;
- c. linguistic processing.

Document representation

1. text processing - a. term selection



a. term selection

- i. Controlled vocabulary: restricted, supervised
- ii. Free-text vocabulary: (almost) entire document

Controlled vocabulary

- A set of well-defined topics/concepts.
- Assigned to documents by humans (or automatically).

Controlled vocabulary

- A set of well-defined topics/concepts.
- Assigned to documents by humans (or automatically).
- Represent documents by **the topics** covered in the document.

Controlled vocabulary

- A set of well-defined topics/concepts.
- Assigned to documents by humans (or automatically).
- Represent documents by **the topics** covered in the document.
- **Which topics to select!**

Controlled

Example - 1

- › Awards and honours
- › Career statistics
 - Personal life
 - Business interests
- › Political career
- › Role in public awareness and philanthropy
- › Post-playing career
 - In media
 - Notes
 - References

one of the greatest [batsman](#) of all time.^[4] He is holder of several world records, including being the all-time highest [run](#)-scorer in cricket,^[6] receiving the most [player of the match](#) awards in international cricket,^[7] and being the only batsman to score 100 international centuries.^[8] Tendulkar was a [Member of Parliament, Rajya Sabha](#) by [presidential nomination](#) from 2012 to 2018.^{[9][10]}

Tendulkar took up cricket at the age of eleven, made his Test match debut on 15 November 1989 against [Pakistan](#) in [Karachi](#) at the age of sixteen, and went on to represent [Mumbai](#) domestically and [India](#) internationally for over 24 years.^[11] In 2002, halfway through his career, *[Wisden](#)* ranked him the second-



Tendulkar in 2017

https://en.wikipedia.org/wiki/Sachin_Tendulkar

Controlled

Example - 1

Yorkshire cricketers | West Zone cricketers | India Blue cricketers | Mumbai Indians cricketers
Indian Premier League coaches | Kerala Blasters FC owners | Recipients of the Bharat Ratna
Honorary Members of the Order of Australia | Recipients of the Padma Vibhushan in sports
Recipients of the Padma Shri in sports | Recipients of the Khel Ratna Award | Recipients of the Arjuna Award
Recipients of the Maharashtra Bhushan Award | Wisden Cricketers of the Year | Wisden Leading Cricketers in the World
International Cricket Council Cricketer of the Year | Man of the Tournament in ODI Worldcup | Indian Air Force officers
Commonwealth Games competitors for India | Cricketers who have acted in films
Nominated members of the Rajya Sabha | People named in the Pandora Papers

https://en.wikipedia.org/wiki/Sachin_Tendulkar

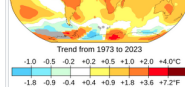
Controlled vocab

Example - 2

- > Society and culture
- > History
- > References
- External links

since the Industrial Revolution.^{[1][4]} Fossil fuel use, deforestation, and some agricultural and industrial practices release greenhouse gases.^[5] These gases absorb some of the heat that the Earth radiates after it warms from sunlight, warming the lower atmosphere. Carbon dioxide, the primary gas driving global warming, has increased in concentration by about 50% since the pre-industrial era to levels not seen for millions of years.^[6]

Climate change has an increasingly large impact on the environment. Deserts are expanding, while heat waves and wildfires are becoming more common.^[7] Amplified warming in the Arctic has contributed to thawing permafrost, retreat of glaciers and sea ice decline.^[8] Higher temperatures are also causing more intense storms, droughts, and other weather extremes.^[9] Rapid environmental change in mountains, coral reefs, and the Arctic is forcing many species to relocate or become extinct.^[10] Even if



Changes in surface air temperature over the past 50 years.^[1] The Arctic has warmed the most, and temperatures on land have generally increased more than sea surface temperatures.



https://en.wikipedia.org/wiki/Climate_change

Controlle

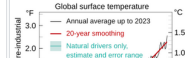
Example - 2

Categories: Climate change | Anthropocene | History of climate variability and change
Global environmental issues | Human impact on the environment

are pre-industrial era to levels not seen for millions of years.

Climate change has an increasingly large [impact on the environment](#). Deserts are expanding, while [heat waves](#) and [wildfires](#) are becoming more common.^[7] [Amplified warming in the Arctic](#) has contributed to thawing permafrost, retreat of glaciers and [sea ice decline](#).^[8] Higher temperatures are also causing [more intense storms](#), droughts, and other [weather extremes](#).^[9] Rapid environmental change in [mountains](#), [coral reefs](#), and the [Arctic](#) is forcing many species to relocate or [become extinct](#).^[10] Even if

Changes in [surface air temperature](#) over the past 50 years.^[11] The [Arctic](#) has warmed the most, and temperatures on land have generally increased more than [sea surface temperatures](#).



https://en.wikipedia.org/wiki/Climate_change

Controlled vocabulary

Features

- Describes topics that are most central to the document.
- The topics do not need to appear explicitly in the text.
- Relationships between concepts facilitate non-query based navigation and exploration.

Controlled vocabulary

Features

- Describes topics that are most central to the document.
- The topics do not need to appear explicitly in the text.
- Relationships between concepts facilitate non-query based navigation and exploration.
- Developed by experts who know the data and the users. (**Expensive**).

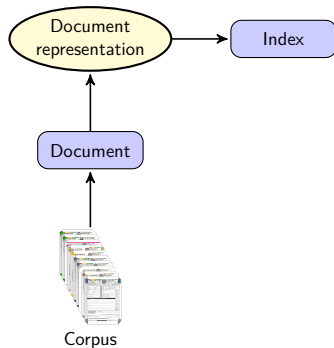
Controlled vocabulary

Features

- Describes topics that are most central to the document.
- The topics do not need to appear explicitly in the text.
- Relationships between concepts facilitate non-query based navigation and exploration.
- Developed by experts who know the data and the users. (**Expensive**).
- Could miss rare but important topics.
- Basic term overlap while retrieval will not work.

Document representation

1. text processing - a. term selection



a. term selection

- i. Controlled vocabulary: restricted, supervised
- ii. Free-text vocabulary: (almost) entire document

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.
- Which terms to select?

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.
- Which terms to select?
 - ▶ most descriptive/important terms only?

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.
- Which terms to select?
 - ▶ most descriptive/important terms only?
 - ▶ most unambiguous terms only?

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.
- Which terms to select?
 - ▶ most descriptive/important terms only?
 - ▶ most unambiguous terms only?
 - ▶ all the terms?

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.
- Which terms to select?
 - ▶ most descriptive/important terms only?
 - ▶ most unambiguous terms only?
 - ▶ all the terms? ✓

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.
- Which terms to select?
 - ▶ most descriptive/important terms only?
 - ▶ most unambiguous terms only?
 - ▶ all the terms? ✓
- Flexibility of expressing higher-level concepts by term combinations while querying.

Document representation:

1. text processing - a. term selection - ii. free-text vocabulary

- Represent documents by **the terms** in the document.
- Which terms to select?
 - ▶ most descriptive/important terms only?
 - ▶ most unambiguous terms only?
 - ▶ all the terms? ✓
- Flexibility of expressing higher-level concepts by term combinations while querying.
- Term importance will be determined during retrieval by the retrieval model.

Free-text vocabulary

```
1 Title: Julius Caesar
2
3 Author: William Shakespeare
4
5 Release Date: November 1998 [eBook #1522]
6 [Most recently updated: November 18, 2021]
7
8 Language: English
9
10
11 Produced by: the PG Shakespeare Team, a team of about twenty Project Gutenberg volunteers.
12
13 *** START OF THE PROJECT GUTENBERG EBOOK JULIUS CAESAR ***
14
15
16
17
18 THE TRAGEDY OF JULIUS CAESAR
19
20
21 by William Shakespeare
22
23
24 Contents
25
26 ACT I
27 Scene I. Rome. A street.
28 Scene II. The same. A public place.
29 Scene III. The same. A street.
30
31
32 ACT II
```

Free-text vocabulary

98 ACT I
99
100 SCENE I. Rome. A street.
101
102 Enter Flavius, Marullus and a throng of Citizens.
103
104 FLAVIUS.
105 Hence! home, you idle creatures, get you home.
106 Is this a holiday? What, know you not,
107 Being mechanical, you ought not walk
108 Upon a labouring day without the sign
109 Of your profession? Speak, what trade art thou?
110
111 CARPENTER.
112 Why, sir, a carpenter.
113
114 MARULLUS.
115 Where is thy leather apron and thy rule?
116 What dost thou with thy best apparel on?
117 You, sir, what trade are you?
118
119 COBBLER.
120 Truly, sir, in respect of a fine workman, I am but, as you would say, a
121 cobbler.
122
123 MARULLUS.
124 But what trade art thou? Answer me directly.
125
126 COBBLER.

Document representation: 1. text processing a. term selection

controlled vocabulary vs. free-text vocabulary

	Cost of assigning index terms	Ambiguity of index terms	Detail of representation
Controlled Vocabularies	High/Low?	Ambiguous/ Unambiguous?	Can represent arbitrary level of detail?
Free-text Vocabularies	High/Low?	Ambiguous/ Unambiguous?	Can represent arbitrary level of detail?

Document representation: 1. text processing a. term selection

controlled vocabulary vs. free-text vocabulary

	Cost of assigning index terms	Ambiguity of index terms	Detail of representation
Controlled Vocabularies	High	Unambiguous	Can't represent arbitrary level of details
Free-text Vocabularies	Low	Can be Ambiguous	Any level of details can be represented

Document representation: 1. text processing a. term selection

controlled vocabulary vs. free-text vocabulary

	Cost of assigning index terms	Ambiguity of index terms	Detail of representation
Controlled Vocabularies	High	Unambiguous	Can't represent arbitrary level of details
Free-text Vocabularies	Low	Can be Ambiguous	Any level of details can be represented

- Free-text vocabulary - cheap and easy (our focus).
- Both are effective and used depending on the need.

Document representation: 1. text processing a. term selection

controlled vocabulary vs. free-text vocabulary

	Cost of assigning index terms	Ambiguity of index terms	Detail of representation
Controlled Vocabularies	High	Unambiguous	Can't represent arbitrary level of details
Free-text Vocabularies	Low	Can be Ambiguous	Any level of details can be represented

- Free-text vocabulary - cheap and easy (our focus).
- Both are effective and used depending on the need.

Document representation:

1. text processing

- 1 Gerard A. "Gerry" Salton (8 March 1927 in Nuremberg – 28 August 1995), was a Professor of Computer Science at Cornell University. Salton was perhaps the leading computer scientist working in the field of information retrieval during his time, and "the father of Information Retrieval". His group at Cornell developed the SMART Information Retrieval System, which he initiated when he was at Harvard. It was the very first system to use the now popular vector space model for Information Retrieval.
- 2 Salton was born Gerhard Anton Sahlmann on March 8, 1927 in Nuremberg, Germany. He received a Bachelor's (1950) and Master's (1952) degree in mathematics from Brooklyn College, and a Ph.D. from Harvard in Applied Mathematics in 1958, the last of Howard Aiken's doctoral students, and taught there until 1965, when he joined Cornell University and co-founded its department of Computer Science.
- 3 Salton was perhaps most well known for developing the now widely used vector space model for Information Retrieval. In this model, both documents and queries are represented as vectors of term counts, and the similarity between a document and a query is given by the cosine between the term vector and the document vector. In this paper, he also introduced TF-IDF, or term-frequency-inverse-document frequency, a model in which the score of a term in a document is the ratio of the number of terms in that document divided by the frequency of the number of documents in which that term occurs. (The concept of inverse document frequency, a measure of specificity, had been introduced in 1972 by Karen Sparck-Jones.) Later in life, he became interested in automatic text summarization and analysis, as well as automatic hypertext generation. He published over 150 research articles and 5 books during his life.
- 4 Salton was editor-in-chief of the Communications of the ACM and the Journal of the ACM, and chaired Special Interest Group on Information Retrieval (SIGIR). He was an associate editor of the ACM Transactions on Information Systems. He was an ACM Fellow (elected 1995), received an Award of Merit from the American Society for Information Science (1989), and was the first recipient of the SIGIR Award for outstanding contributions to study of Information Retrieval (1983) -- now called the Gerard Salton Award.

Document representation:

1. text processing - b. term filtering

Text processing

- a. term selection: select all terms (primary focus).
- b. term filtering;
- c. linguistic processing.

Document representation:

1. text processing - b. term filtering

Text processing

- a. term selection: select all terms (primary focus).
- b. term filtering;
- c. linguistic processing.

term filtering

Document representation:

1. text processing - b. term filtering

Text processing

- a. term selection: select all terms (primary focus).
- b. term filtering;
- c. linguistic processing.

term filtering

- i. tokenization

Document representation:

1. text processing - b. term filtering

Text processing

- a. term selection: select all terms (primary focus).
- b. term filtering;
- c. linguistic processing.

term filtering

- i. tokenization
- ii. down-casing

Document representation:

1. text processing - b. term filtering

Text processing

- a. term selection: select all terms (primary focus).
- b. term filtering;
- c. linguistic processing.

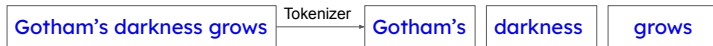
term filtering

- i. tokenization
- ii. down-casing
- iii. stopwords removal

1. text processing - b. term filtering

i. tokenization

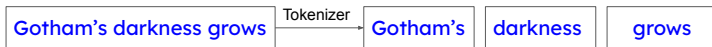
- Tokenization: process to split a string into smaller string objects (**atomic unit of document**).



1. text processing - b. term filtering

i. tokenization

- Tokenization: process to split a string into smaller string objects (**atomic unit of document**).



tokenisation using Python

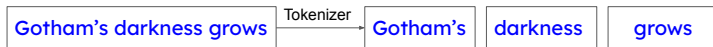
```
import nltk
nltk.download('punkt')
```

```
from nltk import word_tokenize
with open('filename.txt') as fp:
    text = fp.read()
    tokenlist = word_tokenize(text)
```

1. text processing - b. term filtering

i. tokenization

- Tokenization: process to split a string into smaller string objects (**atomic unit of document**).



tokenisation using Python

```
import nltk  
nltk.download('punkt')
```

```
from nltk import word_tokenize  
with open('filename.txt') as fp:  
    text = fp.read()  
    tokenlist = word_tokenize(text)
```

issues with tokenization

- ▶ Ph.D., U.S.A., 3.1415
- ▶ post-covid, non-english
- ▶ Hong Kong
- ▶ O'Connor

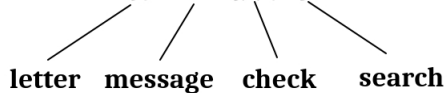
Issues with tokenization

non-english

information retrieval

信息檢索

letter message check search



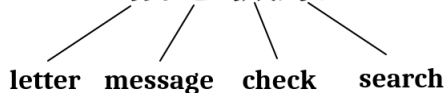
Issues with tokenization

non-english

information retrieval

信息檢索

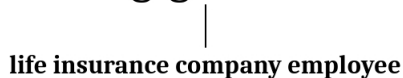
letter message check search



```
graph TD; A[信息檢索] --- B[letter]; A --- C[message]; A --- D[check]; A --- E[search];
```

Lebensversicherungsgesellschaftsangestellter

life insurance company employee



```
graph TD; A[Lebensversicherungsgesellschaftsangestellter] --- B[life insurance company employee];
```

Issues with tokenization

non-english

information retrieval

信息檢索

letter message check search

Lebensversicherungsgesellschaftsangestellter

life insurance company employee

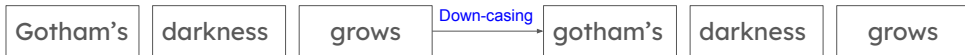
- language detection before tokenization.

1. text processing - b. term filtering

ii. down-casing (a.k.a case-folding)

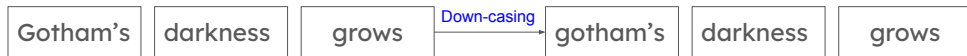
1. text processing - b. term filtering

ii. down-casing (a.k.a case-folding)



1. text processing - b. term filtering

ii. down-casing (a.k.a case-folding)

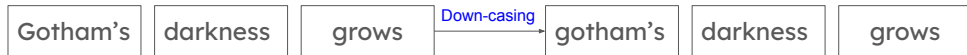


Issues

- Quit beating around the bush
vs.
Former president George Bush

1. text processing - b. term filtering

ii. down-casing (a.k.a case-folding)

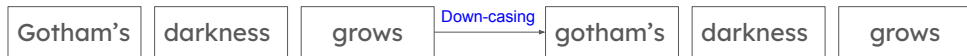


Issues

- Quit beating around the bush
vs.
Former president George Bush
- He lives in US
vs.
It has been done by us

1. text processing - b. term filtering

ii. down-casing (a.k.a case-folding)



Issues

- Quit beating around the bush

vs.

Former president George Bush

- He lives in US

vs.

It has been done by us

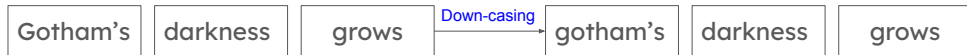
- What happens after cracking C.A.T.?

vs.

The cat was coming after me

1. text processing - b. term filtering

ii. down-casing (a.k.a case-folding)



Issues

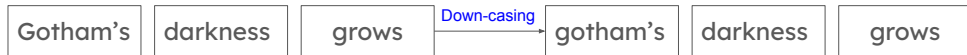
- Quit beating around the bush
vs.
Former president George Bush
- He lives in US
vs.
It has been done by us
- What happens after cracking C.A.T?
vs.
The cat was coming after me

Alternative

- Make some tokens lowercase.
- Simplest heuristic:
 - ▶ lowercase words at the beginning of a sentence;
 - ▶ all words occurring in a title that is all uppercase
 - ▶ in which most or all words are capitalized;
 - ▶ mid-sentence capitalized words left as capitalized.

1. text processing - b. term filtering

ii. down-casing (a.k.a case-folding)



Issues

- Quit beating around the bush
vs.
Former president George Bush
- He lives in US
vs.
It has been done by us
- What happens after cracking C.A.T?
vs.
The cat was coming after me

Alternative

- Make some tokens lowercase.
- Simplest heuristic:
 - ▶ lowercase words at the beginning of a sentence;
 - ▶ all words occurring in a title that is all uppercase
 - ▶ in which most or all words are capitalized;
 - ▶ mid-sentence capitalized words left as capitalized.
- Lowercasing everything often remains the most practical solution.

1. text processing

raw text

A gang of masked criminals rob a mafia-owned bank in Gotham City, betraying and killing each other until the sole survivor, the Joker, reveals himself as the mastermind and escapes with the money. The vigilante Batman, district attorney Harvey Dent, and police lieutenant Jim Gordon ally to eliminate Gotham's organized crime. Batman's true identity, the billionaire Bruce Wayne, publicly supports Dent as Gotham's legitimate protector, believing Dent's success will allow him to retire as Batman and romantically pursue his childhood friend Rachel Dawes—despite her being with Dent.

Gotham's mafia bosses gather to discuss protecting their organizations from the Joker, the police, and Batman. The Joker interrupts the meeting and offers to kill Batman for half of the fortune their accountant, Lau, concealed before fleeing to Hong Kong to avoid extradition. With the help of Wayne Enterprises CEO Lucius Fox, Batman finds Lau in Hong Kong and returns him to the custody of Gotham police.

1. text processing

after tokenization and down-casing

a gang of masked criminals rob a mafia-owned bank in gotham city , betraying and killing each other until the sole survivor , the joker , reveals himself as the master-mind and escapes with the money . the vigilante batman , district attorney harvey dent , and police lieutenant jim gordon ally to eliminate gotham 's organized crime . batman 's true identity , the billionaire bruce wayne , publicly supports dent as gotham 's legitimate protector , believing dent 's success will allow him to retire as batman and romantically pursue his childhood friend rachel dawes—despite her being with dent .

gotham 's mafia bosses gather to discuss protecting their organizations from the joker , the police , and batman . the joker interrupts the meeting and offers to kill batman for half of the fortune their accountant , lau , concealed before fleeing to hong kong to avoid extradition . with the help of wayne enterprises ceo lucius fox , batman finds lau in hong kong and returns him to the custody of gotham police .

1. text processing

after tokenization and down-casing

raw text

A gang of masked criminals rob a mafia-owned bank in Gotham City betraying and killing each other until the sole survivor the Joker reveals himself as the mastermind and escapes with the money

1. text processing

after tokenization and down-casing

raw text

A gang of masked criminals rob a mafia-owned bank in Gotham City betraying and killing each other until the sole survivor the Joker reveals himself as the mastermind and escapes with the money

after down-casing

a gang of masked criminals rob a mafia-owned bank in gotham city betraying and killing each other until the sole survivor the joker reveals himself as the mastermind and escapes with the money

1. text processing

after tokenization and down-casing

raw text

A gang of masked criminals rob a mafia-owned bank in Gotham City betraying and killing each other until the sole survivor the Joker reveals himself as the mastermind and escapes with the money

after down-casing

a gang of masked criminals rob a mafia-owned bank in gotham city betraying and killing each other until the sole survivor the joker reveals himself as the mastermind and escapes with the money

- Popularly referred to as **stopwords**.

1. text processing

after tokenization and down-casing

raw text

A gang of masked criminals rob a mafia-owned bank in Gotham City betraying and killing each other until the sole survivor the Joker reveals himself as the mastermind and escapes with the money

after down-casing

a gang of masked criminals rob a mafia-owned bank in gotham city betraying and killing each other until the sole survivor the joker reveals himself as the mastermind and escapes with the money

- Popularly referred to as **stopwords**.
- Not useful for term-overlapping based retrievals.

Document representation:

1. text processing - b. term filtering

term filtering

- i. tokenization
- ii. down-casing

Document representation:

1. text processing - b. term filtering

term filtering

- i. tokenization
- ii. down-casing
- iii. stopwords removal

Stopword removal

Stopword

- Commonly used words that occur in most of the documents, and do not contribute significantly to the analysis or understanding of the content.

Stopword removal

Stopword

- Commonly used words that occur in most of the documents, and do not contribute significantly to the analysis or understanding of the content.

am is are the a was were at by for with
about into to from any few more
other...

Stopword removal

Stopword

- Commonly used words that occur in most of the documents, and do not contribute significantly to the analysis or understanding of the content.

Issues

- As you may think
- To be or not to be

am is are the a was were at by for with
about into to from any few more
other...

Stopword removal

Stopword

- Commonly used words that occur in most of the documents, and do not contribute significantly to the analysis or understanding of the content.

am is are the a was were at by for with
about into to from any few more
other...

Issues

- As you may think
- To be or not to be
- I was not present at the murder spot

Stopword removal

Stopword

- Commonly used words that occur in most of the documents, and do not contribute significantly to the analysis or understanding of the content.

am is are the a was were at by for with
about into to from any few more
other...

Issues

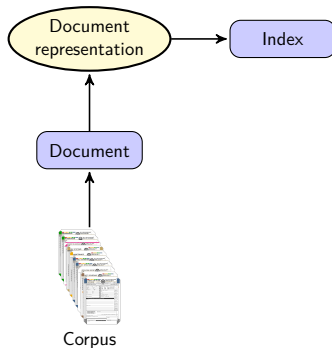
- As you may think
- To be or not to be
- I was not present at the murder spot

Possible Solutions

- 👉 Extensive stopwords removal (200-300 words).
- 👉 Restrictive stopwords removal (10-20 words).
- 👉 No stopwords removal.

Document representation

1. text processing



Document representation

- ☞ text processing
- ☞ indexing

Text processing

- ✓ term selection;
- ✓ term filtering;
- ☞ linguistic processing.

The basics of **searching** in information retrieval

bag of words approach

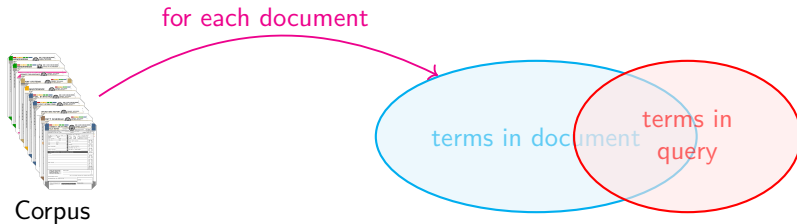


Figure: The bag of words approach to information retrieval.

Retrieval in BOW approach

- Document → **list of keywords**
- User's information need → (natural-language) query → **list of keywords**
- Measure overlap between query and documents.

The basics of **searching** in information retrieval

bag of words approach

- **Document:** containing the term **Indian ... institute ...**

The basics of **searching** in information retrieval

bag of words approach

- **Document:** containing the term **Indian ... institute ...**
- **Query:** **Institution** **in** **India** offering courses on paleontology

The basics of **searching** in information retrieval

bag of words approach

- **Document:** containing the term Indian ... institute ...
- **Query:** Institution in India offering courses on paleontology

institute $\neq$ institution

Text processing

linguistic processing

- Different forms of same words can be used at different places based on the grammar.

Text processing

linguistic processing

- Different forms of same words can be used at different places based on the grammar.
 - ▶ organize, organizes, organizing etc.

Text processing

linguistic processing

- Different forms of same words can be used at different places based on the grammar.
 - ▶ **organize, organizes, organizing** etc.
- Need to have a mechanism to get an unified form.

Text processing

linguistic processing

- Different forms of same words can be used at different places based on the grammar.
 - ▶ **organize, organizes, organizing** etc.
- Need to have a mechanism to get an unified form.
- ① Lemmatization
- ② Stemming

Text processing

linguistic processing

- Different forms of same words can be used at different places based on the grammar.
 - ▶ **organize, organizes, organizing** etc.
- Need to have a mechanism to get an unified form.
- ① Lemmatization
- ② Stemming

Common goal

to reduce inflectional and derivationally related forms of a word to a common base form.

Linguistic processing

lemmatization

Lemmatization

Doing proper reduction to dictionary head words (called **lemma**).

Linguistic processing

lemmatization

Lemmatization

Doing proper reduction to dictionary head words (called **lemma**).

- Reduce inflectional/variant forms to base form
 - ▶ am, are, is $\Rightarrow$ be
 - ▶ car, cars, car's, cars' $\Rightarrow$ car

Linguistic processing

lemmatization

Lemmatization

Doing proper reduction to dictionary head words (called **lemma**).

- Reduce inflectional/variant forms to base form
 - ▶ am, are, is $\Rightarrow$ be
 - ▶ car, cars, car's, cars' $\Rightarrow$ car
- Need to have linguistic knowledge

Linguistic processing

lemmatization

Lemmatization

Doing proper reduction to dictionary head words (called **lemma**).

- Reduce inflectional/variant forms to base form
 - ▶ am, are, is $\Rightarrow$ be
 - ▶ car, cars, car's, cars' $\Rightarrow$ car
- Need to have linguistic knowledge - expensive.

Linguistic processing

stemming

Stemming

Heuristic process of **chopping off ends of words** in the hope of achieving what 'principled' lemmatization attempts to do with a lot of linguistic knowledge.

Linguistic processing

stemming

Stemming

Heuristic process of **chopping off ends of words** in the hope of achieving what 'principled' lemmatization attempts to do with a lot of linguistic knowledge.

institute, institution, institutional

Linguistic processing

stemming

Stemming

Heuristic process of **chopping off ends of words** in the hope of achieving what 'principled' lemmatization attempts to do with a lot of linguistic knowledge.

institute, institution, institutional $\Rightarrow$ institut

Linguistic processing

stemming

Stemming

Heuristic process of **chopping off ends of words** in the hope of achieving what 'principled' lemmatization attempts to do with a lot of linguistic knowledge.

institute, institution, institutional $\Rightarrow$ institut

- Language dependent.

Linguistic processing

stemming

Stemming

Heuristic process of **chopping off ends of words** in the hope of achieving what 'principled' lemmatization attempts to do with a lot of linguistic knowledge.

institute, institution, institutional $\Rightarrow$ **institut**

- Language dependent.
- Common strategy: identify (manually / statistically) set of suffixes and remove from ends of words.

Linguistic processing

stemming

Stemming

Heuristic process of **chopping off ends of words** in the hope of achieving what 'principled' lemmatization attempts to do with a lot of linguistic knowledge.

institute, institution, institutional $\Rightarrow$ institut

- Language dependent.
- Common strategy: identify (manually / statistically) set of suffixes and remove from ends of words.



Linguistic processing

stemming

Stemming

Heuristic process of **chopping off ends of words** in the hope of achieving what 'principled' lemmatization attempts to do with a lot of linguistic knowledge.

institute, institution, institutional $\Rightarrow$ institut

- Language dependent.
- Common strategy: identify (manually / statistically) set of suffixes and remove from ends of words.



- **Stemming** - collapses *derivationally* related words;
- **Lemmatization** - collapses different *inflectional forms* of a lemma.

Stemming

- ① Rule-based stemmer.
- ② Statistical stemmer.

Porter stemmer (rule based stemmer)

- Most common algorithm for stemming English
- 5 phases of reductions
- Phases are applied sequentially
- Each phase consists of a set of commands.

Porter stemmer

few rules

Rule		Example	
SSES	⇒ SS	processes	⇒ process
IES	⇒ I	ponies	⇒ poni
SS	⇒ SS	caress	⇒ caress
S	⇒	cats	⇒ cat

Stemmer vs. Lemmatizer

Input

such an analysis can reveal features that are not easily visible from the variations in the individual genes and can lead to a picture of expression that is more biologically transparent and accessible to interpretation

Stemmer vs. Lemmatizer

Input

such an analysis can reveal features that are not easily visible from the variations in the individual genes and can lead to a picture of expression that is more biologically transparent and accessible to interpretation

Porter stemmer

such an **analysis** can reveal **featur** that ar not **easili visibl** from the **variat** in the **individu gene** and can lead to a **pictur** of express that is more **biolog transpar** and access to **interpret**

Lemmatizer

such an **analysis** can reveal **feature** that are not **easily visible** from the **variation** in the **individual gene** and can lead to a **picture** of expression that is more **biologically transparent** and accessible to **interpretation**

Text processing

till free-text vocabulary...

- Our goal is to describe content using content.
- Are all these words equally descriptive?
- What are the most descriptive words?
- How a computer identify these?

Text processing

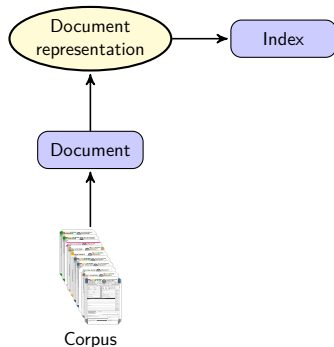
till free-text vocabulary...

- Our goal is to describe content using content.
- Are all these words equally descriptive?
- What are the most descriptive words?
- How a computer identify these?

Will be taken care of by retrieval models.

Document representation

2. storing of terms

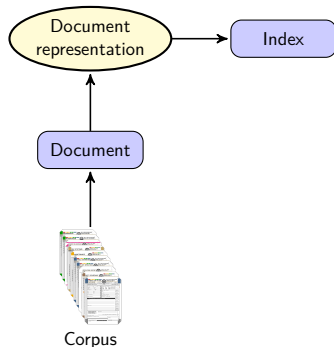


Document representation

- 1 **Text processing:** which terms to select?
 - a. term selection:
 - ★ controlled vocabulary
 - ✓ free-text vocabulary
 - b. term filtering;
 - i. tokenization
 - ii. down-casing
 - iii. stopword removal.
 - c. linguistic processing:
 - ★ lemmatization
 - ✓ stemming

Document representation

2. storing of terms



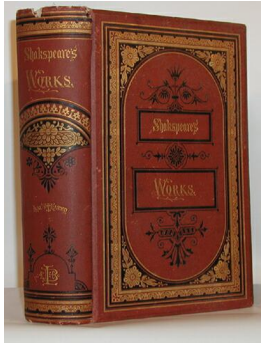
Document representation

1 Text processing: which terms to select?

- a. term selection:
 - ★ controlled vocabulary
 - ✓ free-text vocabulary
- b. term filtering;
 - i. tokenization
 - ii. down-casing
 - iii. stopword removal.
- c. linguistic processing:
 - ★ lemmatization
 - ✓ stemming

2 Indexing: how to store the terms?

A simple IR problem and its possible solutions:



The task

Find the plays containing **Caesar** **AND** **Cleopatra**.

<https://blogger.com>

A real-life IR problem and its possible solution



WIKIPEDIA
The Free Encyclopedia

The task

Find the **Christopher Nolan** movie(s) with **Hugh Jackman** in cast.

A real-life IR pro

Google

Find the Christopher Nolan movie(s) with Hugh Jackman in cast. X |   

All Images Short videos Videos Forums News More ▾ Tools ▾

◆ AI Overview

 Listen

Hugh Jackman has not been in any films with Christopher Nolan as of September 2025. While Jackman has appeared in a variety of genres, including the Christopher Nolan-directed film The Prestige in 2006, Jackman himself is not listed among the cast members of that particular film, and there are no other collaborations between the two artists. 



There appears to be a misunderstanding, as Hugh Jackman starred in The Prestige which was directed by Christopher Nolan. However, this is a common mistake, as there are two actors in The Prestige who have very similar names. It may be helpful to list their correct names and roles to prevent future confusion. 

- **Hugh Jackman:** is not a member of the cast of The Prestige.
- **Hugh Jackman:** is also not a member of any other Nolan-directed films.

The task

Find the **Christopher Nolan** movie(s) with **Hugh Jackman** in cast.

A real-life IR pro

Google Find the Chr

All Images

AI Overv

Listen

Hugh Jack
Nolan as o
a variety of
film The Pr
among the
no other co

There appe
Prestige wh
mistake, as
It may be h
confusion.

- Hugh Ja
- Hugh Ja



Tools



ed in
ed
are

an starred in The
ver, this is a common
e very similar names.
event future

ected films.

Poster source: <https://www.themoviedb.org/>

The task

Find the **Christopher Nolan** movie(s) with **Hugh Jackman** in cast.

A real-life IR pro



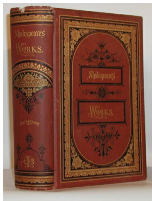
Given a set of terms (query), retrieve documents containing those terms.



The task

Find the **Christopher Nolan** movie(s) with **Hugh Jackman** in cast.

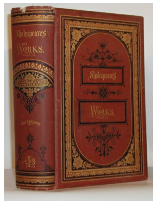
A simple solution



The task

Find plays containing
Caesar **AND** **Cleopatra**.

A simple solution



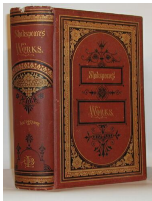
A linear scan

- A linear scan through entire collection for the term.
- For multiple word searching:
Caesar AND Cleopatra:
 - ▶ A linear scan for Term-1(Caesar) $\Rightarrow$ D1
 - ▶ A linear scan for Term-2(Cleopatra) in D1

The task

Find plays containing
Caesar AND Cleopatra.

A simple solution



The task

Find plays containing
Caesar AND Cleopatra.

A linear scan

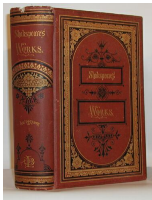
- A linear scan through entire collection for the term.
- For multiple word searching:

Caesar AND Cleopatra:

- ▶ A linear scan for Term-1(**Caesar**) $\Rightarrow$ D1
- ▶ A linear scan for Term-2(**Cleopatra**) in D1

```
grep -l "q1" *.txt
```

A simple solution



A linear scan

- A linear scan through entire collection for the term.
- For multiple word searching:

Caesar AND **Cleopatra**:

- ▶ A linear scan for Term-1(**Caesar**) $\Rightarrow$ D1
- ▶ A linear scan for Term-2(**Cleopatra**) in D1

The task

Find plays containing
Caesar AND **Cleopatra**.

```
grep -l "q1" *.txt
```

Advantage

- Fast if number of documents is small.

Why we need to index the corpus?

Drawback of using grep

- Grep performs a linear search!
- Imagine having 100 thousands documents instead of 100.
- Every search requires a linear search on the whole collection.

Why we need to index the corpus?

Drawback of using grep

- Grep performs a linear search!
- Imagine having 100 thousands documents instead of 100.
- Every search requires a linear search on the whole collection.
- **NOT** type of query (**non trivial**).
- Proximity query (**not possible**).
- **Ranked retrieval** (**Not possible**)

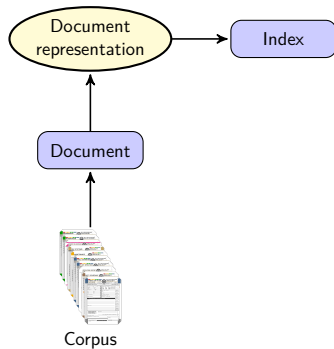
Why we need to index the corpus?

Drawback of using grep

- Grep performs a linear search!
 - Imagine having 100 thousands documents instead of 100.
 - Every search requires a linear search on the whole collection.
 - **NOT** type of query (**non trivial**).
 - Proximity query (**not possible**).
 - **Ranked retrieval** (**Not possible**)
- We need to have an efficient mechanism.

Document representation

2. storing of terms

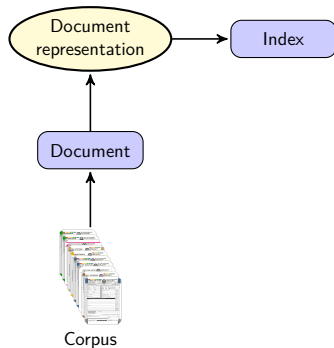


Document representation

- 1 Which terms to select?
- 2 How to store the terms?

Document representation

2. storing of terms



Document representation

- ❶ Which terms to select?
 - ❷ How to store the terms?
-
- 2.1 Term-document incidence matrix.
 - 2.2 Inverted-index.

Term-document incidence matrix

For each document d , *record* all terms t appearing in that document.

Term-document incidence matrix

For each document d , *record* all terms t appearing in that document.

- N - number of documents in the collection.

Term-document incidence matrix

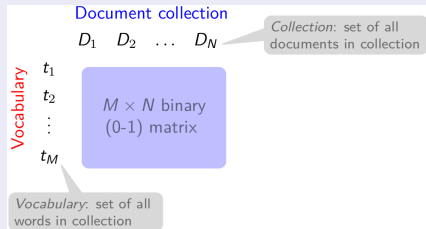
For each document d , *record* all terms t appearing in that document.

- N - number of documents in the collection.
- M - number of *unique* terms in the vocabulary.

Term-document incidence matrix

For each document d , *record* all terms t appearing in that document.

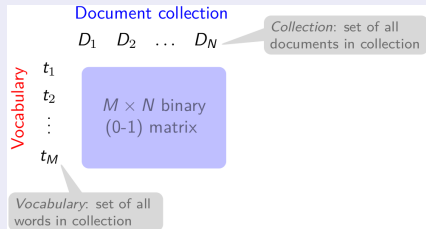
- N - number of documents in the collection.
- M - number of *unique* terms in the vocabulary.
- Make a matrix of size $M \times N$.



Term-document incidence matrix

For each document d , *record* all terms t appearing in that document.

- N - number of documents in the collection.
- M - number of *unique* terms in the vocabulary.
- Make a matrix of size $M \times N$.



- Place term across *row*, and document name identifiers (**docid**) across *column*.

$$\text{matrix}(t, d) = \begin{cases} 1, & \text{if document } d \text{ contains term } t. \\ 0, & \text{if document } d \text{ does not contain term } t. \end{cases}$$

Term-document incidence matrix

- N - num
- M - nu
- Make a

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

- Place term across row, and document name identifiers (**docid**) across column.

$$matrix(t, d) = \begin{cases} 1, & \text{if document } d \text{ contains term } t. \\ 0, & \text{if document } d \text{ does not contain term } t. \end{cases}$$

Term-document incidence matrix

- N - num
- M - nu
- Make a

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0				1	0
sacrifice	0				1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

What happens to the incidence matrix if new documents are added?

- Place term across row, and document name identifiers (**docid**) across column.

$$matrix(t, d) = \begin{cases} 1, & \text{if document } d \text{ contains term } t. \\ 0, & \text{if document } d \text{ does not contain term } t. \end{cases}$$

Incidence vector

- Term-incidence vector: *for each term t , a vector indicating the documents t appears in.*
- Document-incidence vector: *for each document d , a vector indicating the terms occurring in d .*

Incidence vector

- **Term-incidence vector:** *for each term t , a vector indicating the documents t appears in.*
- **Document-incidence vector:** *for each document d , a vector indicating the terms occurring in d .*

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

Term-document incidence matrix

Incidence vector

- **Term-incidence vector:** *for each term t , a vector indicating the documents t appears in.*
- **Document-incidence vector:** *for each document d , a vector indicating the terms occurring in d .*

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

Term-document incidence matrix

`tiv(sacrifice)`

001110:

Incidence vector

- **Term-incidence vector:** *for each term t , a vector indicating the documents t appears in.*
- **Document-incidence vector:** *for each document d , a vector indicating the terms occurring in d .*

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

Term-document incidence matrix

`tiv(sacrifice)`

001110: sacrifice appears in docs 3, 4, 5

Incidence vector

- **Term-incidence vector:** *for each term t , a vector indicating the documents t appears in.*
- **Document-incidence vector:** *for each document d , a vector indicating the terms occurring in d .*

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

Term-document incidence matrix

tiv(sacrifice)

001110: sacrifice appears in docs 3, 4, 5

div(Inception)

0010001:

Incidence vector

- **Term-incidence vector:** *for each term t , a vector indicating the documents t appears in.*
- **Document-incidence vector:** *for each document d , a vector indicating the terms occurring in d .*

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

Term-document incidence matrix

tiv(sacrifice)

001110: sacrifice appears in docs 3, 4, 5

div(Inception)

0010001: Inception has the terms: 3, 7

Incidence vector

- **Term-incidence vector:** for a term t , the vector $tiv(t)$ is a binary vector of length n where $tiv(t)_i = 1$ if t appears in d_i .
 - **Document-incidence vector:** for a document d , the vector $div(d)$ is a binary vector of length m where $div(d)_j = 1$ if d contains term t_j .
- Store IDs for terms and documents.
 - IDs assigned as sequential integers.

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

Term-document incidence matrix

tiv(sacrifice)

001110: sacrifice appears in docs 3, 4, 5

div(Inception)

0010001: Inception has the terms: 3, 7

Term-document incidence matrix

- Place ~~term~~ term identifiers (**termid**) across *row*, and document identifiers (**docid**) across *column*.

$$matrix(t_i, d_j) = \begin{cases} 1, & \text{if document } d_j \text{ contains term with id } t_i. \\ 0, & \text{if document } d_j \text{ does not contain term with id } t_i. \end{cases}$$

Term-document incidence matrix

- Place ~~term~~ term identifiers (**termid**) across *row*, and document identifiers (**docid**) across *column*.

$$\text{matrix}(t_i, d_j) = \begin{cases} 1, & \text{if document } d_j \text{ contains term with id } t_i. \\ 0, & \text{if document } d_j \text{ does not contain term with id } t_i. \end{cases}$$

	d_1	d_2	d_3	d_4	d_5	d_6
t_1	0	1	0	0	1	0
t_2	0	1	0	0	1	1
t_3	1	0	0	0	0	1
t_4	0	1	0	0	1	0
t_5	0	0	1	1	1	0
t_6	0	0	1	1	0	0
t_7	1	0	1	1	0	1

Term-document incidence matrix

- Place ~~term~~ term identifiers (**termid**) across *row*, and document identifiers (**docid**) across *column*.

$$\text{matrix}(t_i, d_j) = \begin{cases} 1, & \text{if document } d_j \text{ contains term with id } t_i. \\ 0, & \text{if document } d_j \text{ does not contain term with id } t_i. \end{cases}$$

	d_1	d_2	d_3	d_4	d_5	d_6
t_1	0	1	0	0	1	0
t_2	0	1	0	0	1	1
t_3	1	0	0	0	0	1
t_4	0	1	0	0	1	0
t_5	0	0	1	1	1	0
t_6	0	0	1	1	0	0
t_7	1	0	1	1	0	1

termid mapping	
t_1	duality
t_2	identity
t_3	memory
t_4	rivalry
t_5	sacrifice
t_6	survival
t_7	time

docid mapping	
d_1	Inception
d_2	The Dark Knight
d_3	Interstellar
d_4	Dunkirk
d_5	The Prestige
d_6	Memento

Query processing

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

survival AND time AND NOT rivalry

- `tiv(survival) AND tiv(time) AND NOT tiv(rivalry)`

Query processing

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

survival AND time AND NOT rivalry

- `tiv(survival) AND tiv(time) AND NOT tiv(rivalry)`
- Do a bitwise AND.

0 0 1 1 0 0

Query processing

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

survival AND time AND NOT rivalry

- `tiv(survival)` AND `tiv(time)` AND NOT `tiv(rivalry)`
- Do a bitwise AND.

```
0 0 1 1 0 0
1 0 1 1 0 1
```

Query processing

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

survival AND time AND NOT rivalry

- $\text{tiv}(\text{survival}) \text{ AND } \text{tiv}(\text{time}) \text{ AND NOT } \text{tiv}(\text{rivalry})$
- Do a bitwise AND.

0	0	1	1	0	0
1	0	1	1	0	1
1	0	1	1	0	1

Query processing

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

survival AND time AND NOT rivalry

- $\text{tiv}(\text{survival}) \text{ AND } \text{tiv}(\text{time}) \text{ AND NOT } \text{tiv}(\text{rivalry})$
- Do a bitwise AND.

0	0	1	1	0	0
1	0	1	1	0	1
1	0	1	1	0	1
1	0	1	1	0	0

Query processing

	Inception	The Dark Knight	Interstellar	Dunkirk	The Prestige	Memento
duality	0	1	0	0	1	0
identity	0	1	0	0	1	1
memory	1	0	0	0	0	1
rivalry	0	1	0	0	1	0
sacrifice	0	0	1	1	1	0
survival	0	0	1	1	0	0
time	1	0	1	1	0	1

survival AND time AND NOT rivalry

- $\text{tiv}(\text{survival}) \text{ AND } \text{tiv}(\text{time}) \text{ AND NOT } \text{tiv}(\text{rivalry})$
- Do a bitwise AND.

0	0	1	1	0	0
1	0	1	1	0	1
1	0	1	1	0	1
1	0	1	1	0	0

Boolean retrieval

Boolean retrieval

query representation

Boolean query assumption

- information need can be represented using boolean constraints: **AND**, **OR**, and **AND NOT**.

Boolean retrieval

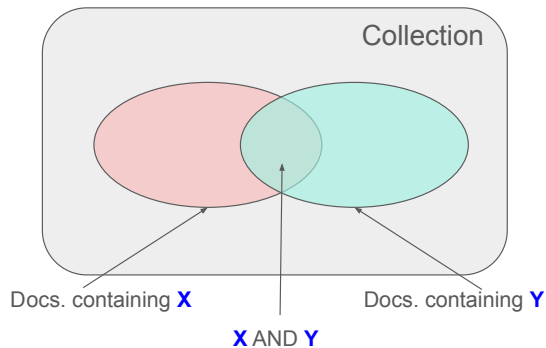
query representation

Boolean query assumption

- information need can be represented using boolean constraints: **AND**, **OR**, and **AND NOT**.
- survival
- survival **AND** time
- survival **AND** (time **OR** rivalry)
- survival **AND** (time **OR** rivalry) **AND NOT** sacrifice

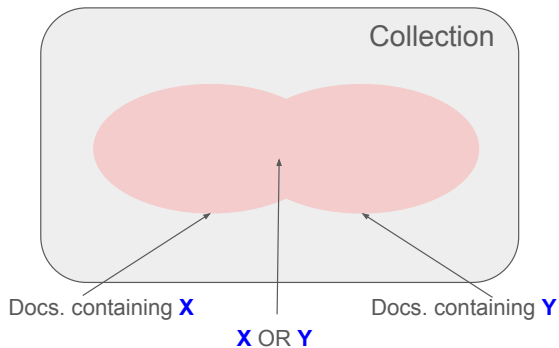
Boolean retrieval

- Query: X AND Y



Boolean retrieval

- Query: $X \text{ OR } Y$



Boolean retrieval

Advantages

- Easy for the system (no ambiguity in the query)
 - ▶ the burden is on the user to formulate the right query.
- The user gets transparency and control:
 - ▶ lots of results $\Rightarrow$ the query is too broad.
 - ▶ no results $\Rightarrow$ the query is too narrow.

Boolean retrieval

Advantages

- Easy for the system (no ambiguity in the query)
 - ▶ the burden is on the user to formulate the right query.
- The user gets transparency and control:
 - ▶ lots of results $\Rightarrow$ the query is too broad.
 - ▶ no results $\Rightarrow$ the query is too narrow.
- Common strategy for finding the right balance:
 - ▶ if the query is too broad, add **AND** or **AND NOT** constraints.
 - ▶ if the query is too narrow, add **OR** constraints.

Boolean retrieval

Advantages

- Easy for the system (no ambiguity in the query)
 - ▶ the burden is on the user to formulate the right query.
- The user gets transparency and control:
 - ▶ lots of results $\Rightarrow$ the query is too broad.
 - ▶ no results $\Rightarrow$ the query is too narrow.
- Common strategy for finding the right balance:
 - ▶ if the query is too broad, add **AND** or **AND NOT** constraints.
 - ▶ if the query is too narrow, add **OR** constraints.

Disadvantages

- All terms considered equally important.
- Hard to formulate query.

Problem: bigger corpus

- Consider:
 - ▶ 1M (1,000,000) documents, each with about 1K (1,000) terms.
 - ▶ Total 1B (1,000,000,000) terms.
 - ▶ Average 6bytes/term including spaces and punctuation:
 - ★ Total approx 6GB (5722MB) of data.
 - ▶ Say, there are 500K distinct terms among these.

Problem: bigger corpus

- Consider:
 - ▶ 1M (1,000,000) documents, each with about 1K (1,000) terms.
 - ▶ Total 1B (1,000,000,000) terms.
 - ▶ Average 6bytes/term including spaces and punctuation:
 - ★ Total approx 6GB (5722MB) of data.
 - ▶ Say, there are 500K distinct terms among these.

Can't build Incidence Matrix!

- ▶ Number of Rows: 500K
- ▶ Number of Columns: 1M

Problem: bigger corpus

- Consider:
 - ▶ 1M (1,000,000) documents, each with about 1K (1,000) terms.
 - ▶ Total 1B (1,000,000,000) terms.
 - ▶ Average 6bytes/term including spaces and punctuation:
 - ★ Total approx 6GB (5722MB) of data.
 - ▶ Say, there are 500K distinct terms among these.

Can't build Incidence Matrix!

- ▶ Number of Rows: 500K
- ▶ Number of Columns: 1M
- ▶ 500K X 1M matrix: half-a-trillion 0's & 1's.

Problem: bigger corpus

- Consider:
 - ▶ 1M (1,000,000) documents, each with about 1K (1,000) terms.
 - ▶ Total 1B (1,000,000,000) terms.
 - ▶ Average 6bytes/term including spaces and punctuation:
 - ★ Total approx 6GB (5722MB) of data.
 - ▶ Say, there are 500K distinct terms among these.

Can't build Incidence Matrix!

- ▶ Number of Rows: 500K
- ▶ Number of Columns: 1M
- ▶ 500K X 1M matrix: half-a-trillion 0's & 1's.
- ▶ But extremely sparse.

Problem: bigger corpus

- Consider:
 - ▶ 1M (1,000,000) documents, each with about 1K (1,000) terms.
 - ▶ Total 1B (1,000,000,000) terms.
 - ▶ Average 6bytes/term including spaces and punctuation:
 - ★ Total approx 6GB (5722MB) of data.
 - ▶ Say, there are 500K distinct terms among these.

Can't build Incidence Matrix!

- ▶ Number of Rows: 500K
- ▶ Number of Columns: 1M
- ▶ 500K X 1M matrix: half-a-trillion 0's & 1's.
- ▶ But extremely sparse. (record the 1 positions?)

Problem: bigger corpus

- Consider:
 - ▶ 1M (1,000,000) documents, each with about 1K (1,000) terms.
 - ▶ Total 1B (1,000,000,000) terms.
 - ▶ Average 6bytes/term including spaces and punctuation:
 - ★ Total approx 6GB (5722MB) of data.
 - ▶ Say, there are 500K distinct terms among these.

Can't build Incidence Matrix!

- ▶ Number of Rows: 500K
- ▶ Number of Columns: 1M
- ▶ 500K X 1M matrix: half-a-trillion 0's & 1's.
- ▶ But extremely sparse. (record the 1 positions?)

- Proximity query
- **Ranked retrieval**

Solution: using an inverted index

Solution: using an inverted index

most important data structure underlying modern retrieval systems

From Incidence Matrix...

Terms	DocIDs											
	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	D11	
Alfred	0	1	0	1	0	0	1	1	0	0	0	
Arthur	1	1	0	0	1	0	0	0	1	0	1	
Batman	0	1	0	0	0	0	0	1	0	0	0	
Bruce	0	1	0	1	0	0	0	1	0	0	1	
⋮												

- Place term/**termid** across *row*, and document identifiers (**docid**) across *column*.

$$\text{matrix}(t_i, d_j) = \begin{cases} 1, & \text{if document } d_j \text{ contains term with id } t_i. \\ 0, & \text{if document } d_j \text{ does not contain term with id } t_i. \end{cases}$$

... to Inverted Index

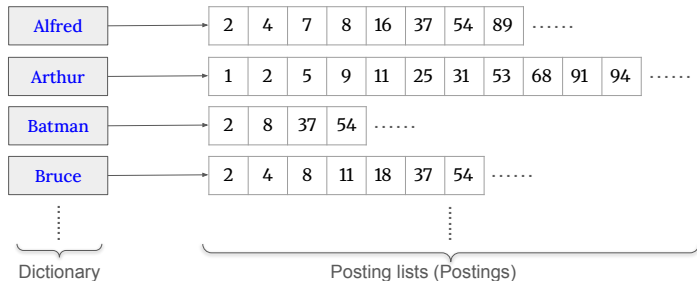
- For each term t , store a list of all documents that contain t .

... to Inverted Index

- For each term t , store a list of all documents that contain t .
 - ▶ identify each doc. by **docid** (e.g. integer starting from 0 or 1).

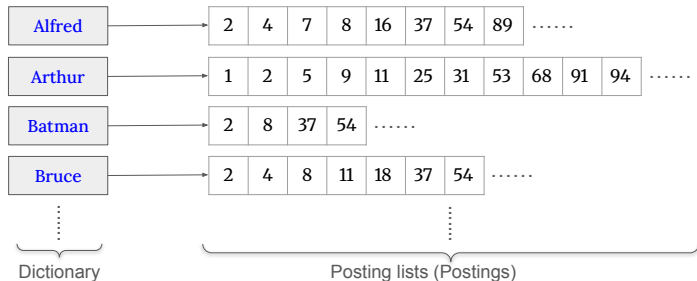
... to Inverted Index

- For each term t , store a list of all documents that contain t .
 - ▶ identify each doc. by **docid** (e.g. integer starting from 0 or 1).



... to Inverted Index

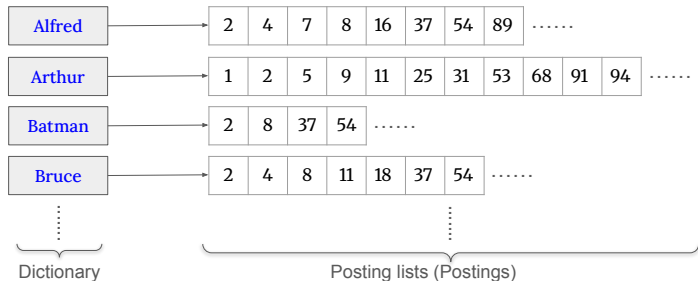
- For each term t , store a list of all documents that contain t .
 - ▶ identify each doc. by **docid** (e.g. integer starting from 0 or 1).



- To store:
 - 1 Dictionary - list.
 - 2 Postings - list per-term.

... to Inverted Index

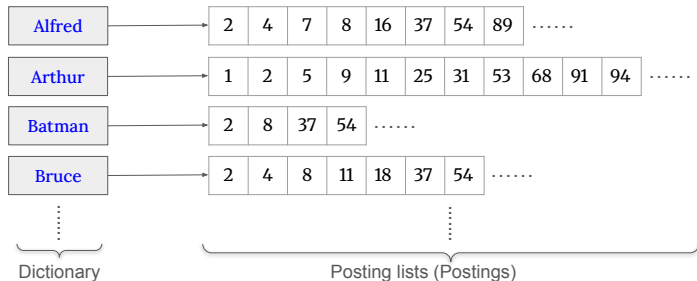
- For each term t , store a list of all documents that contain t .
 - ▶ identify each doc. by **docid** (e.g. integer starting from 0 or 1).



- To store:
 - 1 Dictionary - list.
 - 2 Postings - list per-term.
- Dictionary - lexicographically sorted.
 - ▶ why?
- Posting - sorted by docid.
 - ▶ why?

... to Inverted Index

- For each term t , store a list of all documents that contain t .
 - ▶ identify each doc. by **docid** (e.g. integer starting from 0 or 1).



- To store:
 - 1 Dictionary - list.
 - 2 Postings - list per-term.
- Dictionary - lexicographically sorted.
 - ▶ why?
- Posting - sorted by docid.
 - ▶ why?

Why is it called *Inverted* Index?

Why is it called *Inverted* Index?

- **Forward index:**

- ▶ **the list of documents**, and the terms appearing in them.

DocID		Content
1	→	Bruce loves programming
2	→	Diana loves travelling
3	→	programming is fun
4	→	Diana learns programming

Why is it called *Inverted* Index?

- **Forward index:**

- ▶ **the list of documents**, and the terms appearing in them.

DocID		Content
1	→	Bruce loves programming
2	→	Diana loves travelling
3	→	programming is fun
4	→	Diana learns programming

- **Inverted index:**

- ▶ **the list of terms**, and the documents in which they appear.

Terms		List of docids
Bruce	→	[1]
Diana	→	[2, 4]
is	→	[3]
fun	→	[3]
learns	→	[4]
loves	→	[1, 2]
programming	→	[1, 3, 4]
travelling	→	[2]

Why is it called *Inverted* Index?

- **Forward index:**

- ▶ the list of documents, and the terms appearing in them.

DocID		Content
1	→	Bruce loves programming
2	→	Diana loves travelling
3	→	programming is fun
4	→	Diana learns programming

- **Inverted index:**

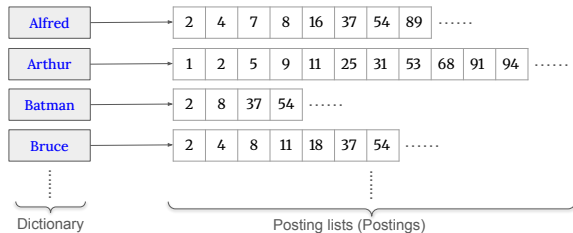
- ▶ the list of terms, and the documents in which they appear.

Terms		List of docids
Bruce	→	[1]
Diana	→	[2, 4]
is	→	[3]
fun	→	[3]
learns	→	[4]
loves	→	[1, 2]
programming	→	[1, 3, 4]
travelling	→	[2]

Forward index: document → terms mapping

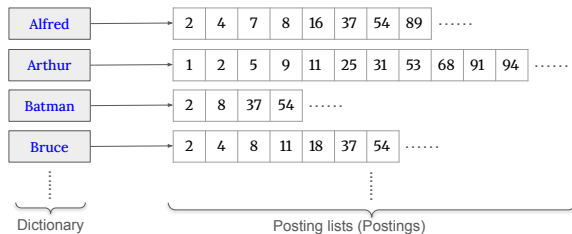
Inverted index: term → documents mapping

Query Processing



Alfred **AND** Bruce

Query Processing



Alfred AND Bruce

- Locate **Alfred** in Dictionary;
 - ▶ Retrieve its postings.
- Locate **Bruce** in Dictionary;
 - ▶ Retrieve its postings.
- *Merge* the two postings.



How to merge



How to merge



- Walk through the two postings simultaneously.

How to merge



- Walk through the two postings simultaneously.
- Report the same docIds.

How to merge



- Walk through the two postings simultaneously.
- Report the same docIds.

Results in:

2

4

8

37

54

How to merge



- Walk through the two postings simultaneously.
- Report the same docIds.

Results in:

2	4	8	37	54
---	---	---	----	----

Note:

- If length of postings are x and y , merge takes $O(x+y)$ operations.

How to merge



- Walk through the two postings simultaneously.
- Report the same doclds.

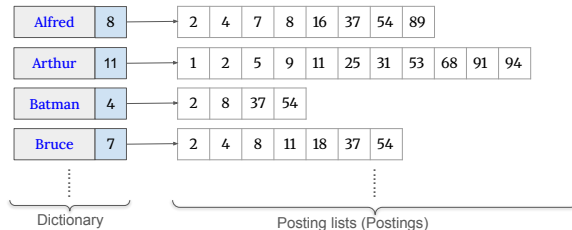
Results in: 2 4 8 37 54

Note:

- If length of postings are x and y , merge takes $O(x+y)$ operations.
- **Postings must be sorted by docld.**

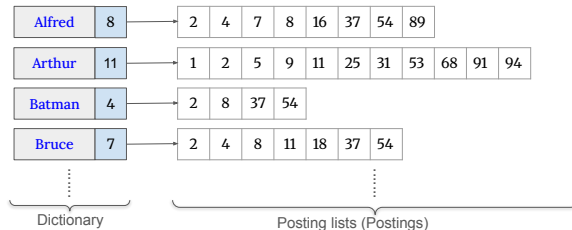
Query optimization

- Consider **document frequency (df)** saved along with the term in dictionary.



Query optimization

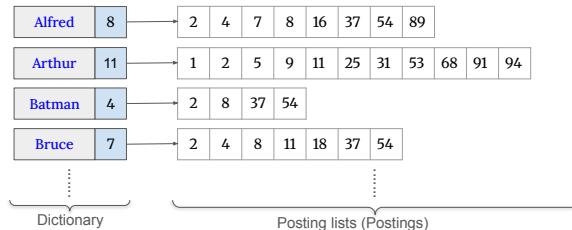
- Consider **document frequency** (**df**) saved along with the term in dictionary.



- Consider a query which is **AND** of t terms.

Query optimization

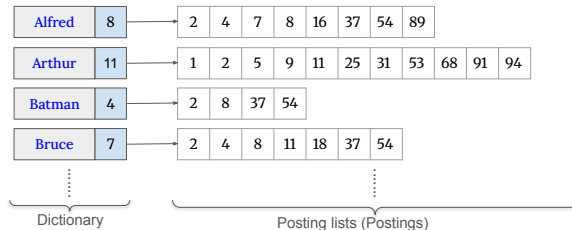
- Consider **document frequency** (**df**) saved along with the term in dictionary.



- Consider a query which is **AND** of t terms.
- For each of the t terms, get their postings; **AND** them together.

Query optimization

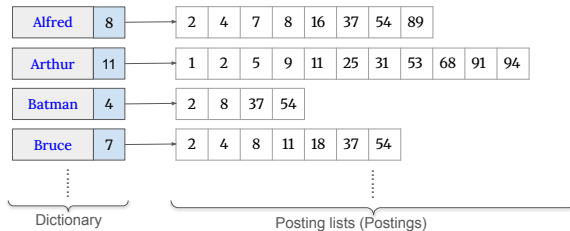
- Consider **document frequency** (**df**) saved along with the term in dictionary.



- Consider a query which is AND of t terms.
- For each of the t terms, get their postings; AND them together.
- What is the best order for query processing?

Query optimization: continued

- Process in the order of increasing frequency.
 - ▶ Start with the smallest set; then keep cutting further.



Query:

Execute the query as:

(Batman AND Bruce) AND Alfred ...

(a bit) Complex query optimization

Query:

(Alfred OR Arthur) AND (Batman OR Bruce)

- Get frequency for all terms.
- Estimate size of each *OR*ed pair by the sum of term's frequency.
- Process in increasing order of *OR* sizes.

(a bit) Complex query optimization

Query:

(Alfred OR Arthur) AND (Batman OR Bruce)

- Get frequency for all terms.
- Estimate size of each *OR*ed pair by the sum of term's frequency.
- Process in increasing order of *OR* sizes.

possible because we are storing the **document frequency** of the terms.

(a bit) Complex query optimization

Query:

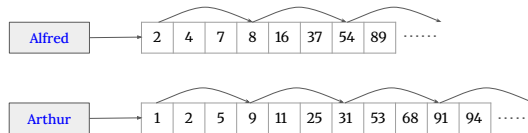
(Alfred OR Arthur) AND (Batman OR Bruce)

- Get frequency for all terms.
- Estimate size of each *OR*ed pair by the sum of term's frequency.
- Process in increasing order of *OR* sizes.

possible because we are storing the **document frequency** of the terms.
more on **document frequency** later.

Faster merging

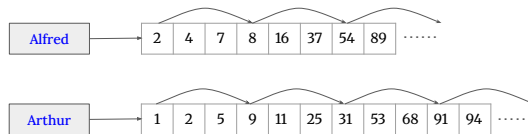
using skip pointers



- Shortcuts to avoid processing parts of postings list.

Faster merging

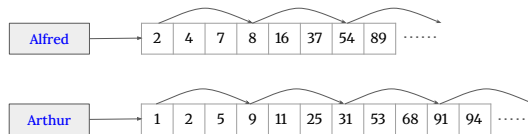
using skip pointers



- Shortcuts to avoid processing parts of postings list.
- The two obvious questions:
 - ① where to place skip pointers and
 - ② how to do efficient merging using skip pointers.

Faster merging

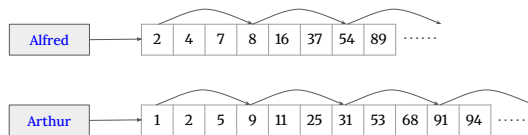
using skip pointers



- Shortcuts to avoid processing parts of postings list.
- The two obvious questions:
 - ① where to place skip pointers and
 - ② how to do efficient merging using skip pointers.
- More skips → shorter skip spans;

Faster merging

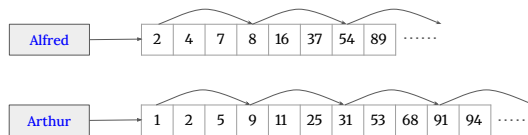
using skip pointers



- Shortcuts to avoid processing parts of postings list.
- The two obvious questions:
 - ① where to place skip pointers and
 - ② how to do efficient merging using skip pointers.
- More skips → shorter skip spans; lots of comparisons to skip pointers, lots of space storing pointers.

Faster merging

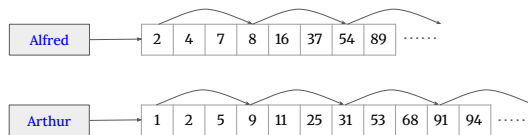
using skip pointers



- Shortcuts to avoid processing parts of postings list.
- The two obvious questions:
 - ① where to place skip pointers and
 - ② how to do efficient merging using skip pointers.
- More skips → shorter skip spans; lots of comparisons to skip pointers, lots of space storing pointers.
- Fewer skips → few pointer comparisons;

Faster merging

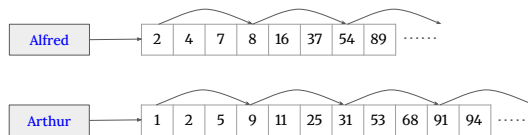
using skip pointers



- Shortcuts to avoid processing parts of postings list.
- The two obvious questions:
 - ① where to place skip pointers and
 - ② how to do efficient merging using skip pointers.
- More skips → shorter skip spans; lots of comparisons to skip pointers, lots of space storing pointers.
- Fewer skips → few pointer comparisons; there will be fewer opportunities to skip.

Faster merging

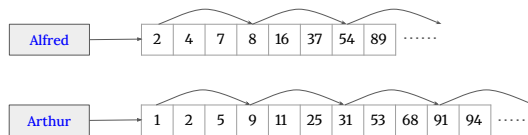
using skip pointers



- Shortcuts to avoid processing parts of postings list.
- The two obvious questions:
 - ① where to place skip pointers and
 - ② how to do efficient merging using skip pointers.
- More skips $\rightarrow$ shorter skip spans; lots of comparisons to skip pointers, lots of space storing pointers.
- Fewer skips $\rightarrow$ few pointer comparisons; there will be fewer opportunities to skip.
- For a postings list of length P use $\sqrt{P}$ evenly-spaced skip pointers $\rightarrow$ found to work well in practice.

Faster merging

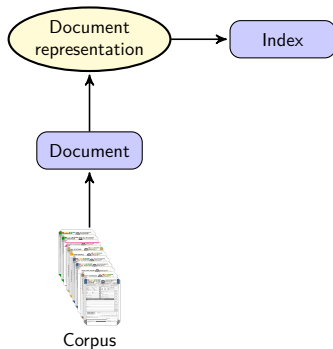
using skip pointers



- Shortcuts to avoid processing parts of postings list.
- The two obvious questions:
 - ① where to place skip pointers and
 - ② how to do efficient merging using skip pointers.
- More skips $\rightarrow$ shorter skip spans; lots of comparisons to skip pointers, lots of space storing pointers.
- Fewer skips $\rightarrow$ few pointer comparisons; there will be fewer opportunities to skip.
- For a postings list of length P use $\sqrt{P}$ evenly-spaced skip pointers $\rightarrow$ found to work well in practice.
- **Iff the index is static.**

Document representation

1. text processing - putting everything together

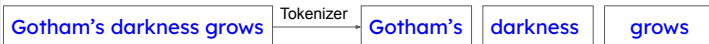


Document representation - 1. text processing

- a term selection:
 - i. controlled vocabulary
 - ii. free-text vocabulary
- b term filtering:
 - i. tokenization
 - ii. case-folding
 - iii. stopword removal
- c linguistic processing:
 - i. lemmatization
 - ii. stemming

Inverted index construction - putting everything together

Tokenization: separating individual terms

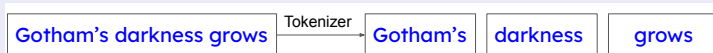


Linguistic processing: each terms into its *root* word



Inverted index construction - putting everything together

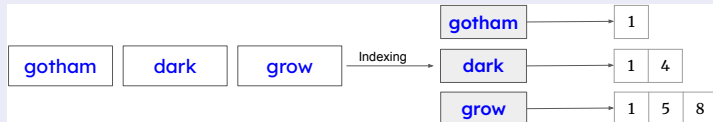
Tokenization: separating individual terms



Linguistic processing: each terms into its *root* word



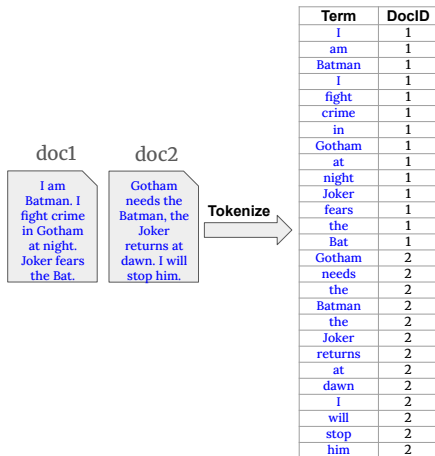
Indexing: index all terms



Indexing

token sequencing

- Make sequence of (processed-token, docid) pairs.



Indexing

sort

- Sort the terms:
 - 1 First lexicographically.
 - 2 Then by docid.

Term	DocID
I	1
am	1
Batman	1
I	1
fight	1
crime	1
in	1
Gotham	1
at	1
night	1
Joker	1
fears	1
the	1
Bat	1
Gotham	2
needs	2
the	2
Batman	2
The	2
Joker	2
returns	2
at	2
dawn	2
I	2
will	2
stop	2
him	2

Sorting
→

Term	DocID
am	1
at	1
at	2
Bat	1
Batman	1
Batman	2
crime	1
dawn	2
fears	1
fight	1
Gotham	1
Gotham	2
him	2
I	1
I	1
I	2
in	1
Joker	1
Joker	2
needs	2
night	1
returns	2
stop	2
the	1
the	2
the	2
will	2

Indexing

Dictionary and postings

- Merge multiple term entries in a single document.

Term	DocID
am	1
at	1
at	2
Bat	1
Batman	1
Batman	2
crime	1
dawn	2
fears	1
fight	1
Gotham	1
Gotham	2
him	2
I	1
I	1
I	2
in	1
Joker	1
Joker	2
needs	2
night	1
returns	2
stop	2
the	1
the	2
the	2
will	2

Combine

Term	DocID	freq
am	1	1
at	1	1
at	2	1
Bat	1	1
Batman	1	1
Batman	2	1
crime	1	1
dawn	2	1
fears	1	1
fight	1	1
Gotham	1	1
Gotham	2	1
him	2	1
I	1	2
I	2	1
in	1	1
Joker	1	1
Joker	2	1
needs	2	1
night	1	1
returns	2	1
stop	2	1
the	1	1
the	2	2
will	2	1

Indexing

Dictionary and posting (cont.)

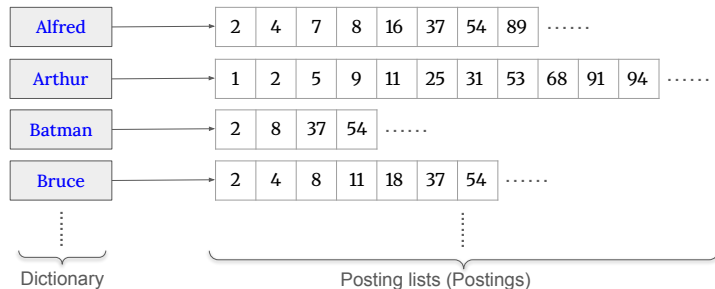
- Split into dictionary (with frequency) and posting.

Term	DocID	freq
am	1	1
at	1	1
at	2	1
Bat	1	1
Batman	1	1
Batman	2	1
crime	1	1
dawn	2	1
fears	1	1
fight	1	1
Gotham	1	1
Gotham	2	1
him	2	1
I	1	2
I	2	1
in	1	1
Joker	1	1
Joker	2	1
needs	2	1
night	1	1
returns	2	1
stop	2	1
the	1	1
the	2	2
will	2	1

Combine

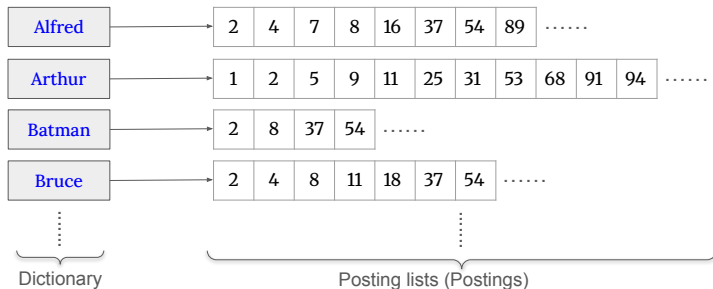
Term	DF	CF	DocID
am	1	1	1
at	2	2	1, 2
Bat	1	1	1
Batman	2	2	1, 2
crime	1	1	1
dawn	1	1	2
fears	1	1	1
fight	1	1	1
Gotham	2	2	1, 2
him	1	1	2
I	2	3	1, 2
in	1	1	1
Joker	2	2	1, 2
needs	1	1	2
night	1	1	1
returns	1	1	2
stop	1	1	2
the	2	3	1, 2
will	1	1	2

Things to be stored in an inverted index



- 1 **Dictionary:** containing all the terms of the collection (less frequently updated).

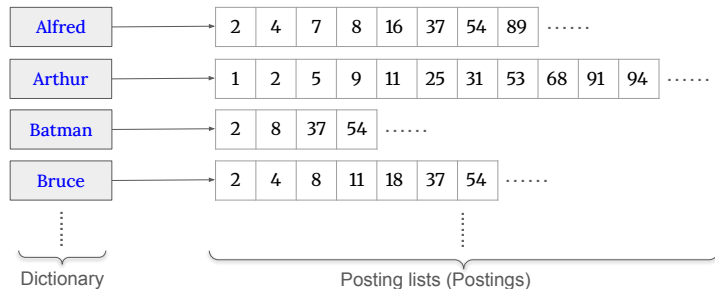
Things to be stored in an inverted index



❶ **Dictionary:** containing all the terms of the collection (less frequently updated).

- ▶ the actual term;
- ▶ document frequency;

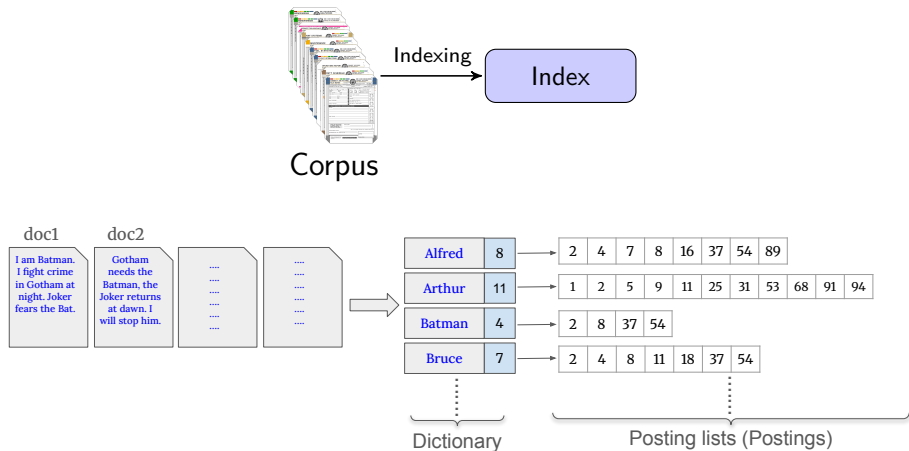
Things to be stored in an inverted index



- 1 **Dictionary**: containing all the terms of the collection (less frequently updated).
 - ▶ the actual term;
 - ▶ document frequency;
 - ▶ some information regarding the start of the posting list of the term.
- 2 **Posting list**: for each term, containing the doclds in which it appears (frequently updated).

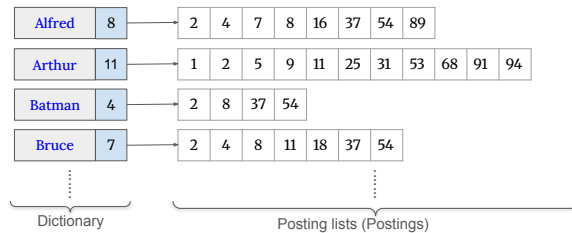
Information Retrieval

indexing



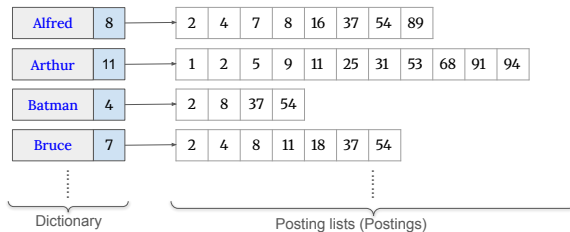
Information Retrieval

retrieval



Information Retrieval

retrieval



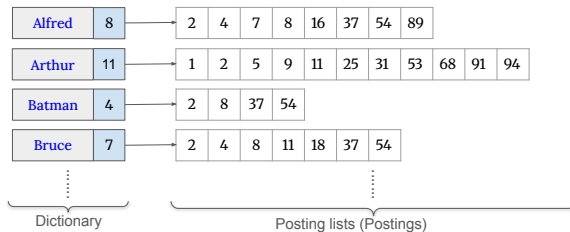
Query

- Arthur Batman

Arthur Batman → 2

Information Retrieval

retrieval



Query

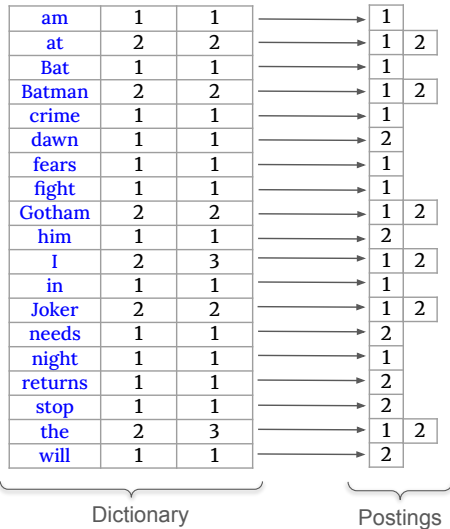
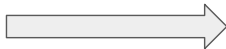
- Arthur Batman

Arthur Batman → 2

still boolean retrieval!

Why still stuck with Boolean retrieval?

Term	DF	CF	DocID
am	1	1	1
at	2	2	1, 2
Bat	1	1	1
Batman	2	2	1, 2
crime	1	1	1
dawn	1	1	2
fears	1	1	1
fight	1	1	1
Gotham	2	2	1, 2
him	1	1	2
I	2	2	1, 2
in	1	1	1
Joker	2	2	1, 2
needs	1	1	2
night	1	1	1
returns	1	1	2
stop	1	1	2
the	2	3	1, 2
will	1	1	2



Why still stuck with Boolean retrieval?

Term	DF	CF	DocID
am	1	1	1
at	2	2	1, 2
Bat	1	1	1
Batman	2	2	1, 2
crime	1	1	1
dawn	1	1	2
fears	1	1	1
fight	1	1	1
Gotham	2	2	1, 2
him	1	1	2
I	2	2	1, 2
in	1	1	1
Joker	2	2	1, 2
needs	1	1	2
night	1	1	1
returns	1	1	2
stop	1	1	2
the	2	3	1, 2
will	1	1	2

- Only recording whether a term is present in the document or not.

am	1	1	→	1	
at	2	2	→	1	2
Bat	1	1	→	1	
Batman	2	2	→	1	2
crime	1	1	→	1	
dawn	1	1	→	2	
fears	1	1	→	1	
fight	1	1	→	1	
Gotham	2	2	→	1	2
him	1	1	→	2	
I	3	3	→	1	2
in	1	1	→	1	
Joker	2	2	→	1	2
needs	1	1	→	2	
night	1	1	→	1	
returns	1	1	→	2	
stop	1	1	→	2	
the	2	3	→	1	2
will	1	1	→	2	

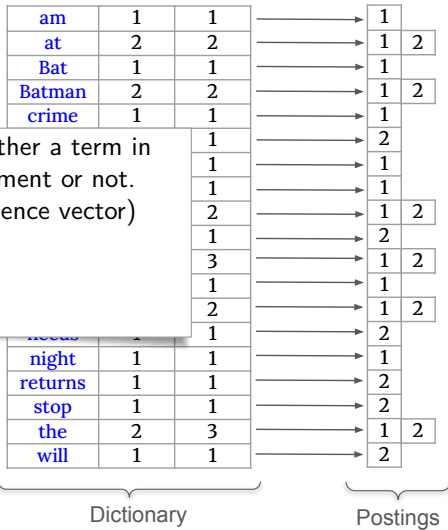
Dictionary

Postings

Why still stuck with Boolean retrieval?

Term	DF	CF	DocID
am	1	1	1
at	2	2	1, 2
Bat	1	1	1
Batman	2	2	1, 2
crime	1	1	1
dawn	1	1	2
fears	1	1	1
fight	1	1	1
Gotham	2	2	1, 2
him	1	1	2
I	2	2	1, 2
in	1	1	1
Joker	2	2	1, 2
needs	1	1	2
night	1	1	1
returns	1	1	2
stop	1	1	2
the	2	3	1, 2
will	1	1	2

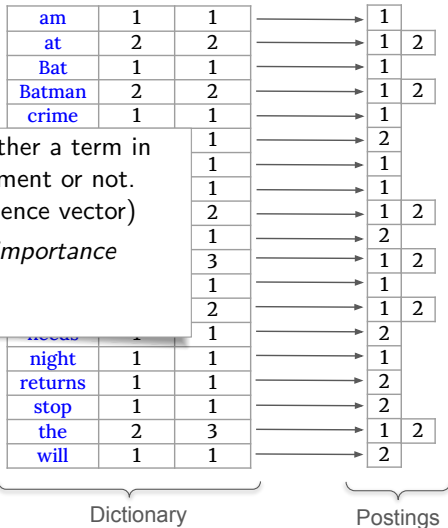
- Only recording whether a term is present in the document or not. (same as term incidence vector)



Why still stuck with Boolean retrieval?

Term	DF	CF	DocID
am	1	1	1
at	2	2	1, 2
Bat	1	1	1
Batman	2	2	1, 2
crime	1	1	1
dawn	1	1	2
fears	1	1	1
fight	1	1	1
Gotham	2	2	1, 2
him	1	1	2
I	2	2	1, 2
in	1	1	1
Joker	2	2	1, 2
needs	1	1	2
night	1	1	1
returns	1	1	2
stop	1	1	2
the	2	3	1, 2
will	1	1	2

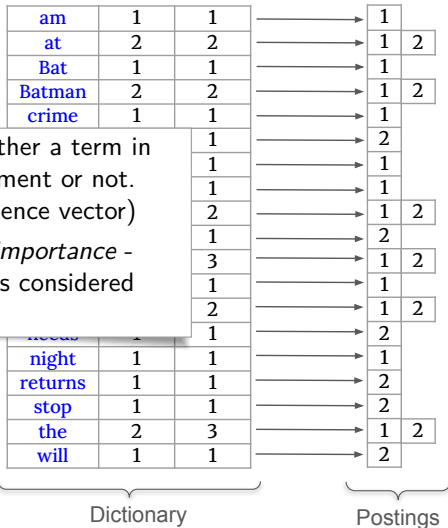
- Only recording whether a term is present in the document or not. (same as term incidence vector)
- No notion of *term importance*



Why still stuck with Boolean retrieval?

Term	DF	CF	DocID
am	1	1	1
at	2	2	1, 2
Bat	1	1	1
Batman	2	2	1, 2
crime	1	1	1
dawn	1	1	2
fears	1	1	1
fight	1	1	1
Gotham	2	2	1, 2
him	1	1	2
I	2	2	1, 2
in	1	1	1
Joker	2	2	1, 2
needs	1	1	2
night	1	1	1
returns	1	1	2
stop	1	1	2
the	2	3	1, 2
will	1	1	2

- Only recording whether a term is present in the document or not. (same as term incidence vector)
- No notion of *term importance* - presence of all terms considered equally important.



Indexing beyond Boolean Retrieval

Term	DF	CF	DocID
am	1	1	1
at	2	2	1, 2
Bat	1	1	1
Batman	2	2	1, 2
crime	1	1	1
dawn	1	1	2
fears	1	1	1
fight	1	1	1
Gotham	2	2	1, 2
him	1	1	2
I	2	3	1, 2
in	1	1	1
Joker	2	2	1, 2
needs	1	1	2
night	1	1	1
returns	1	1	2
stop	1	1	2
the	2	3	1, 2
will	1	1	2

